01 — WHO I AM

Stefano Carlo *Lambertenghi.*

I like taking systems apart to see why they break, then putting them back together better. Fortuna, the research vehicle, is the biggest thing I have taken apart so far.

Researcher at fortiss, doctoral candidate at TUM, and technical lead for Fortuna, our autonomous-driving research vehicle.



THE SHORT VERSION

Background.

I'm Stefano, an Italian computer scientist and researcher currently living in Munich. I grew up in Sondrio, a small town in the Italian Alps, and studied computer science in Verona, Italy, before moving to Lugano, Switzerland, to study artificial intelligence at USI.

While at USI, I met my current PhD supervisor, Andrea Stocco, at Formula USI, a hackathon he organized around the Donkey Car platform. That little car ended up being more important than I expected. When Andrea later offered a PhD position working on autonomous driving and testing, I followed him to Germany. We still use the same Donkey Car platform in our research today.

Before moving to Munich, my main project at IDSIA was my Master's thesis on autonomous nano-drones, which I continued for another year as an intern. I worked on fitting a relative-localization neural network onto a 40-gram quadrotor, allowing it to track another drone in real time using a single onboard camera and less than 100 mW of power. The work eventually became a paper at IEEE ICRA 2023.

In 2023, I joined fortiss and the Technical University of Munich for my PhD. The robots became considerably larger. Today most of my work revolves around autonomous driving and, in particular, understanding what happens when systems tested in simulation are deployed in the real world.

That has made my PhD much more hands-on than I originally expected. Alongside experimental design and research papers, I spend a lot of time working with ROS, sensors, vehicle interfaces, computers, cables, and whatever has decided not to work that morning. I am also the technical lead for Fortuna, our full-size autonomous research vehicle.

There is probably a common thread through all of this. I like working on systems where software eventually has to interact with something physical. I'm interested in machine learning and simulation, but mostly in what happens when you actually deploy them and discover which assumptions were wrong.

Outside research, I simply enjoy playing with hardware and building things. Fortuna's recovery started as work and gradually became a bit of a hobby. Some of the less serious things I have built around it ended up in the Just for fun section.

Growing up in the Alps also means I have been skiing for as long as I can remember. When there is no snow, I enjoy windsurfing. I also like travelling by motorcycle with my girlfriend, Francesca. Moving to Munich gave me a sufficiently good excuse to buy a BMW GS, which has since become our preferred way of disappearing somewhere for a few days.

CURRENT	Researcher at fortiss · Doctoral researcher at TUM
FOCUS	Testing and validation of autonomous systems
METHODS	Simulation, mixed reality, field experiments, empirical software engineering
PLATFORMS	Fortuna (full-size) and a small-scale Donkey Car research vehicle

EMAIL ME ↗ GITHUB ↗ VIEW CV ↗

02 — RESEARCH / SIMULATION ↔ REALITY

Research.

I study how well results transfer between simulation and the real world, how to make an autonomous-driving system more robust once that gap is measured, and how to reuse real driving data to make simulation more realistic.



RESEARCH THEMES

Six research themes.

01

Reality-gap measurement

Measure which actuation, perception, and behavioral differences matter when a test moves between virtual and physical environments.

Found: Vehicle-in-the-loop testing reduced the actuation gap, and mixed-reality testing reduced the perception gap, compared with simulation alone.

ASE · 2025
A Multi-Modality Evaluation of the Reality Gap in Autonomous Driving Systems ↗

02

Robustness under perturbation

Stress camera and LiDAR pipelines, then follow the effect beyond model accuracy into timing, planning, control, and system behavior.

Found: Offline robustness was only weakly predictive of closed-loop failures on the real vehicle; training on perturbed data improved robustness under adverse conditions.

ICST · 2025
Benchmarking Image Perturbations for Testing Automated Driving Assistance Systems ↗

SCIENCE OF COMPUTER PROGRAMMING · 2026
PerturbationDrive: A Framework for Perturbation-Based Testing of ADAS ↗

03

Focused failure discovery

Use near misses and predicted future states to spend the testing budget where failures are more likely to surface.

Found: Focusing the search around forecasted near misses found failures using a smaller testing budget than an unfocused search.

ICSE · 2026
Misbehavior Forecasting for Focused Autonomous Driving Systems Testing ↗

04

Real-to-simulation reconstruction

Turn recorded road scenes into replayable simulations that preserve the geometry, trajectory, and visual character of the original drive.

Found: It's not a full replacement for the real environment, but it lets us reuse real drives for closed-loop testing.

ASE · TOOLS & DATASETS · 2026
Cam2Sim: Neural Scenario Reconstruction for Closed-Loop Autonomous Driving Simulation ↗

ICST · 2024
Assessing Quality Metrics for Neural Reality Gap Input Mitigation in Autonomous Driving Testing ↗

05

Execution environments

The interfaces and tooling that let the same test run in simulation, hardware-in-the-loop, or on a physical vehicle, without rewriting it each time.

ASE · 2025
A Multi-Modality Evaluation of the Reality Gap in Autonomous Driving Systems ↗

06

Research platforms

Fortuna, the small-scale platform, and the tools around them are maintained so other students and researchers can keep using them, not just demo them once.

VEHICLE
Fortuna, the research platform ↗

STUDENTS
Teaching and student projects ↗

SEEING IT WORK

Demos.

Papers describe the method. These recordings show what it actually looks like.



CAM2SIM

Real drives, replayed in simulation.

Cam2Sim started from a practical problem. We had many hours of real driving recordings, but once an experiment was finished we could only replay the recorded sensor data. We wanted to turn these recordings into environments in which the driving system could interact again. Cam2Sim reconstructs the recorded environment in CARLA and uses Gaussian Splatting for the camera view. The reconstruction is not a complete replacement for the real environment, but it allows us to reuse real experiments for closed-loop simulation.

READ THE PAPER ↗

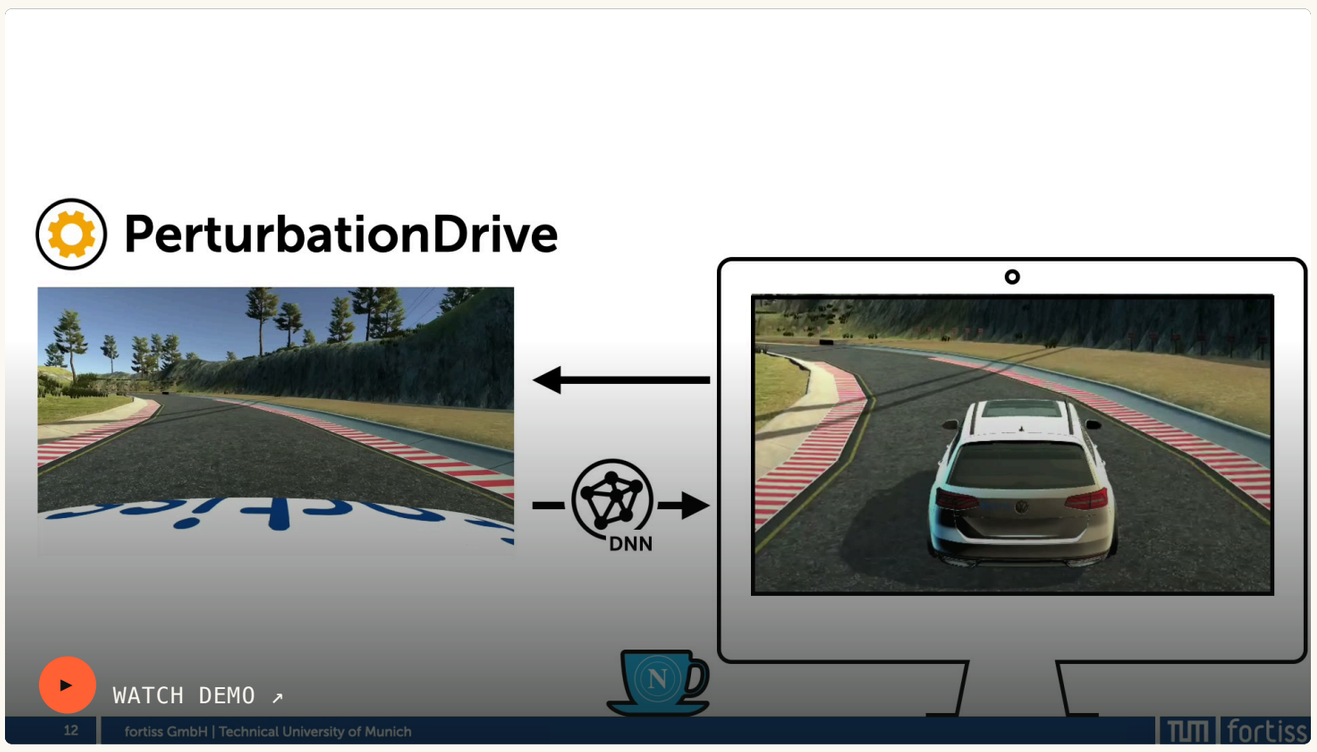


ASE 2025

The same reality gap, across three execution levels.

An open-source small-scale vehicle and its Unity digital twin, compared against Fortuna at full scale, across simulation, vehicle-in-the-loop, and mixed-reality testing.

READ THE PAPER ↗

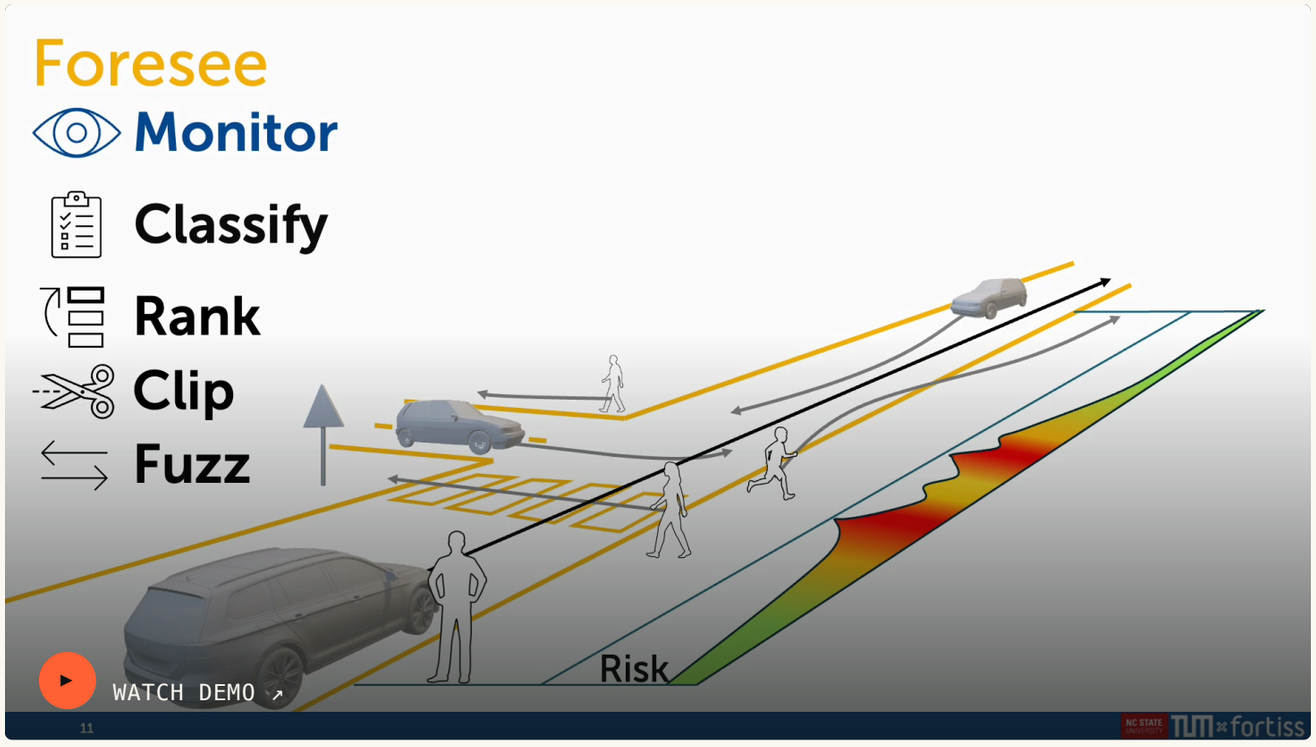


PERTURBATIONDRIVE

The same perturbation, any simulator.

PerturbationDrive generates camera perturbations—like rain, or a corrupted feed—and applies them through a simulator adapter, so the same scenario runs unchanged across Udacity's simulator, a small-scale digital twin, or a full driving stack. This recording walks through the scenario generator and simulator adapter behind the ICST 2025 benchmark.

READ THE PAPER ↗



FORESEE

Forecasting near misses to focus fuzzing.

Foresee monitors nearby agents, classifies and ranks the riskiest ones, then clips and fuzzes the scenario around them—so testing time goes where failures are most likely to surface.

READ THE PAPER ↗

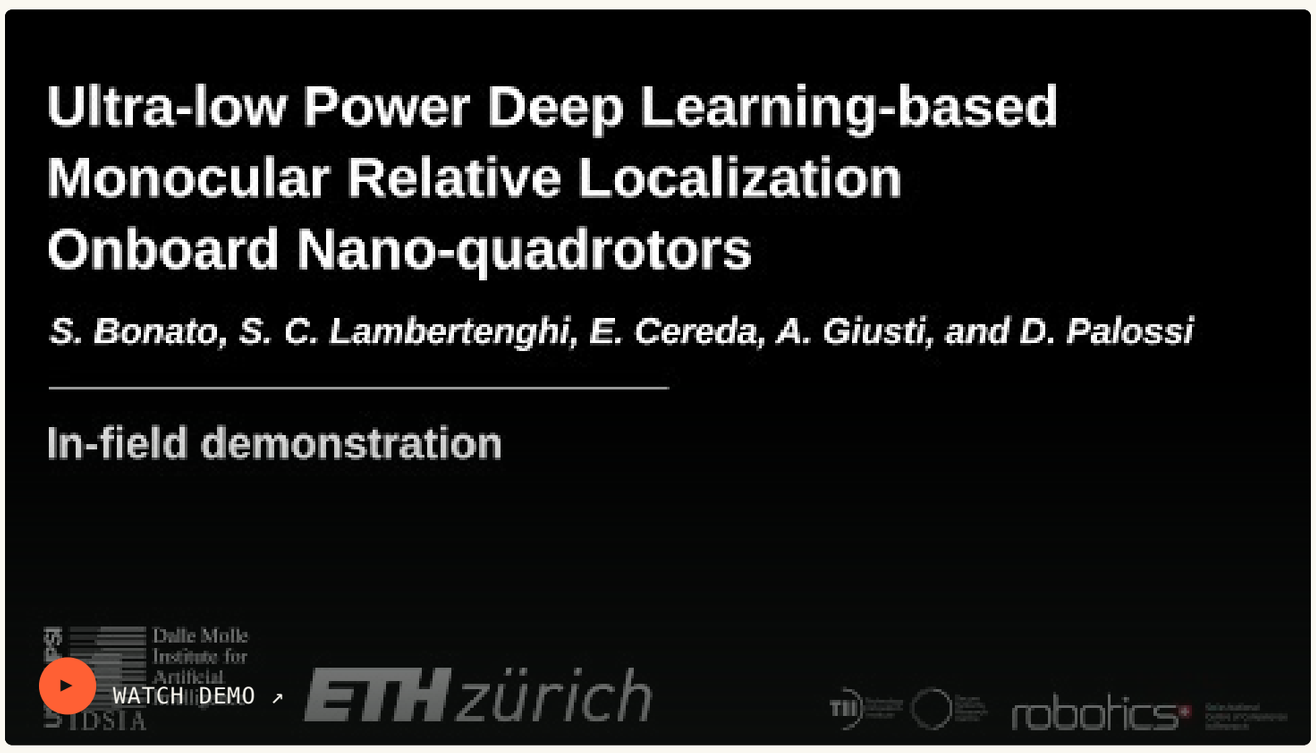


ICST 2024

Do image-quality metrics track what actually matters?

A recording of the ICST 2024 talk on assessing quality metrics for neural reality-gap input mitigation—testing whether standard image-quality metrics track how neural image translation affects a driving system's perception and control.

READ THE PAPER ↗



BEFORE THE REALITY GAP

Fitting a neural network onto a 40-gram nano-drone.

Before autonomous cars, I worked on autonomous nano-drones. At the Dalle Molle Institute for Artificial Intelligence (IDSIA) in Lugano, I helped fit a relative-localization neural network onto a 40-gram nano-quadrotor: a single onboard camera and under 100 mW of power, enough to track a peer drone in real time (IEEE ICRA 2023). The platform changed; the question didn't: what has to be true for a learned model to work outside the lab it was trained in?

READ THE PAPER ↗

RESEARCH IN THE OPEN

A result is more valuable when someone else can interrogate it. The work is accompanied wherever possible by **code, datasets, replication packages, and executable examples**—from PerturbationDrive to Cam2Sim and vehicle-level artifacts.

BROWSE OPEN-SOURCE WORK ↗

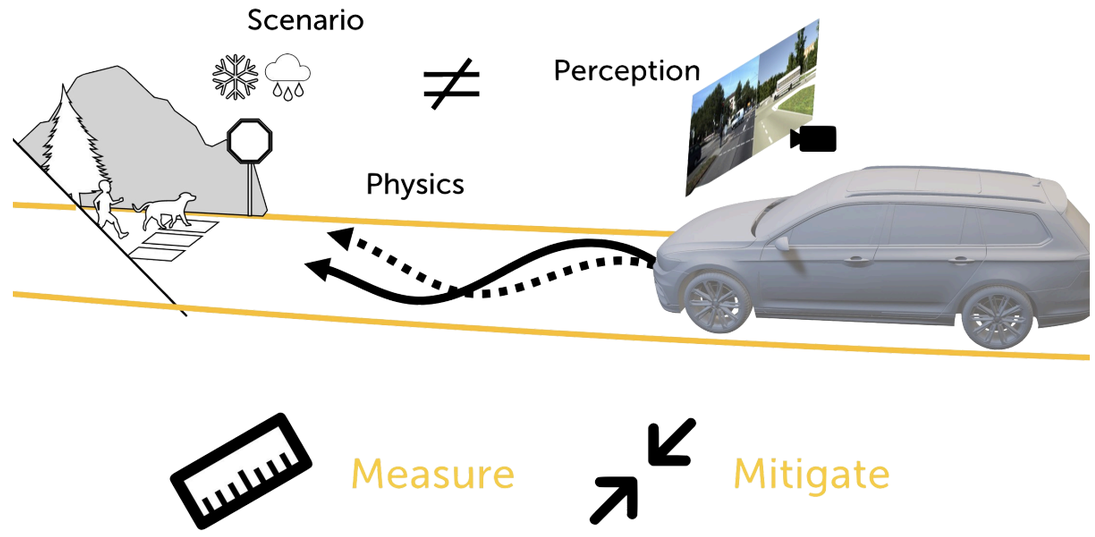
READ THE PAPERS ↗

03 — 2023–2026

Selected publications.

Peer-reviewed papers, executable tools, and replication packages on the reality gap, robustness under perturbation, and reusing real driving data for simulation. A complete list is available on DBLP.

DBLP PROFILE ↗



2026 ASE

Real-World Perturbation Testing of Autonomous Driving Systems

Stefano Carlo Lambertenghi, Matthias Weil, Andrea Stocco

We wanted to know whether robustness results measured offline predict what happens when the same perturbations are applied to an autonomous vehicle. We first tested the models offline and then repeated the experiments on Fortuna. Across more than 740 experiments, the relationship was weaker than we expected. We also tested whether fine-tuning on perturbed data improved real driving performance.

PAPER ↗ ARTIFACT ↗

2026 ICSE

Misbehavior Forecasting for Focused Autonomous Driving Systems Testing

M. M. Abid Naziri, Stefano Carlo Lambertenghi, Andrea Stocco, Marcelo d'Amorim

We wanted to know whether we could predict which scenarios are likely to expose failures before running a full test suite. We built Foresee, which monitors nearby agents, ranks near misses by risk, and clips and fuzzes the scenario around them. Focusing the search this way found failures using a smaller testing budget than an unfocused search.

PAPER ↗

2026 ASE · TOOLS & DATASETS

Cam2Sim: Neural Scenario Reconstruction for Closed-Loop Autonomous Driving Simulation

Davide Jannussi, Stefano Carlo Lambertenghi, Constantin Carste, Andrea Stocco

We had many hours of real driving recordings, but once an experiment was finished we could only replay the recorded sensor data. Cam2Sim reconstructs the recorded environment in CARLA and uses Gaussian Splatting for the camera view, so we can reuse real experiments for closed-loop simulation.

PAPER ↗ ARTIFACT ↗

2026 SCIENCE OF COMPUTER PROGRAMMING

PerturbationDrive: A Framework for Perturbation-Based Testing of ADAS

Hannes Leonhard, Stefano Carlo Lambertenghi, Andrea Stocco

PerturbationDrive started as the benchmark behind our ICST 2025 paper, but we wanted it to work as reusable infrastructure rather than a one-off study. This paper describes the resulting framework: more than 30 image perturbations, offline evaluation on static datasets, and online closed-loop testing across multiple simulators, combined with procedural road generation and search-based testing.

PAPER ↗ ARTIFACT ↗

2025 ASE

A Multi-Modality Evaluation of the Reality Gap in Autonomous Driving Systems

Stefano Carlo Lambertenghi, Mirena Flores Valdez, Andrea Stocco

We wanted to know how much an autonomous driving system's behavior changes as it moves from simulation to a real vehicle, and whether an intermediate setup closes part of that gap. We compared simulation, vehicle-in-the-loop, mixed-reality, and real-world testing on a small-scale vehicle and its digital twin, for both end-to-end and modular models. Vehicle-in-the-loop reduced the actuation gap, and mixed-reality testing reduced the perception gap compared to simulation alone.

PAPER ↗ ARTIFACT ↗

2025 ICST

Benchmarking Image Perturbations for Testing Automated Driving Assistance Systems

Stefano Carlo Lambertenghi, Hannes Leonhard, Andrea Stocco

We wanted to know which kinds of image perturbations expose weaknesses in automated-driving perception and control, and whether training on perturbed images helps. We benchmarked 38 perturbation types across camera-based driving systems. Training on perturbed images improved robustness under adverse conditions.

IEEE COMPUTER SOCIETY TCSE DISTINGUISHED PAPER AWARD

PAPER ↗ ARTIFACT ↗

2024 ICST

Assessing Quality Metrics for Neural Reality Gap Input Mitigation in Autonomous Driving Testing

Stefano Carlo Lambertenghi, Andrea Stocco

We wanted to know whether standard image-quality metrics track how neural image translation affects a driving system, rather than just how it looks to a human. We tested this across vehicle detection and end-to-end lane keeping using several metrics and translation methods. The metrics did not track downstream performance consistently across both tasks, which argues for using them carefully as a proxy.

PAPER ↗ ARTIFACT ↗

2023 ICRA

Ultra-Low Power Deep Learning-based Monocular Relative Localization Onboard Nano-Quadrotors

Stefano Bonato, Stefano Carlo Lambertenghi, Elia Cereda, Alessandro Giusti, Daniele Palossi

We wanted to know whether two nano-drones could localize each other in real time using only a single onboard camera, within the power budget of a 40-gram quadrotor. We trained a small neural network for this and deployed it onboard. It ran at 48 Hz using about 95 mW and reached a mean localization error of about 15 cm in flight tests.

PAPER ↗

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04 — FORTUNA / FULL-SCALE RESEARCH PLATFORM

A street-legal *research vehicle.*



When I started working with Fortuna in 2023, the vehicle had been inactive for more than a year. I took responsibility for bringing it back into operation. This involved reconstructing parts of the existing architecture and operating procedures, restoring the software and hardware components needed for autonomous driving, and gradually making the vehicle usable for research again.

Today I lead the technical work around the vehicle. We use it for research on autonomous-driving testing, simulation, and real-world validation. Several students and researchers contribute to the platform and use it for their own work.

2018 VW PASSAT VARIANT GTE | CAMERAS + 3 LIDARS | DRIVE-BY-WIRE | ROS

2023
RECOVERY AND REINTEGRATION BEGAN

48h+
REAL-WORLD DRIVING IN THE 2026 PERTURBATION STUDY

Multiple
MODULAR AND END-TO-END ADS ARCHITECTURES

2
VEHICLE SCALES—A FULL-SIZE FORTUNA AND A SMALL-SCALE DONKEY CAR

INSIDE THE VEHICLE

Two ways to drive the same *car*.

The driving stack can be an end-to-end model or a modular perception-planning-control pipeline. Bring your own laptop or PC, up to 1000 W, connect it over Ethernet, and start driving.



END-TO-END MODEL — CAMERA IN, STEERING AND THROTTLE OUT.



MODULAR PIPELINE — PERCEPTION, PLANNING, AND CONTROL AS SEPARATE STAGES.

THE PLATFORM

Fortuna is fortiss's full-size autonomous-driving research vehicle: a retrofitted, street-legal Volkswagen Passat Variant GTE with five cameras, three LiDARs providing 360° coverage, precise localization, onboard compute, and real-time access to steering and acceleration through the vehicle gateway.

We use it to run the same research question across offline models, hardware-in-the-loop setups, and a complete moving vehicle, then compare where the answer changes.

INHERITED, RECOVERED, CHANGED

What I actually did.

01

Inherited

Drive-by-wire actuation and control, the camera and LiDAR hardware, and the Apollo integration with GPS/IMU and the CAN bus—built for Providentia between 2016 and 2021.

02

Recovered

In 2023, after more than a year of inactivity: reassessed every subsystem, restored communication paths, and made the vehicle inspectable and operable again.

03

Changed since 2023

A ROS bridge alongside Apollo so any laptop or PC can drive the car over Ethernet, modular and end-to-end driving stacks, simulation integration, hardware-in-the-loop infrastructure, and the small-scale Donkey Car platform.

RECOVERY → RESEARCH

From recovery to research.

Fortuna wasn't built for me. It was fortiss's automated-driving prototype for Providentia—then the project ended, and the car sat unused.

2016–21

Built for Providentia

As fortiss's 5G-connected automated-vehicle prototype, Fortuna gathered data alongside roadside sensor infrastructure for Providentia, a project researching connected, automated driving on Germany's A9 digital motorway test bed near Munich. When the project ended in 2021, so did the car's job.

2023

Understand and recover

I assessed each inactive subsystem, restored communication paths, and made the vehicle inspectable and operable again.

2024–25

Rebuild it modular

Rather than restore the original single-purpose setup, I rebuilt Fortuna as a modular, ROS-based platform: the driving stack can simply be swapped, and so can the compute behind it—any laptop or PC up to 1000 W, connected over Ethernet. With students joining the project, we integrated modular and end-to-end driving stacks, sensor pipelines, and simulation.

2026

Test at full scale

For our ASE 2026 study, I compared 72 camera and LiDAR perturbations across offline analysis, hardware-in-the-loop execution, and closed-loop driving on Fortuna: more than 740 experiments and over 48 hours of autonomous driving.

NEXT

Turn fieldwork into shared infrastructure

I keep connecting physical drives to reproducible scenarios, open tools, student projects, and research questions the next team can build on.

SYSTEM ANATOMY

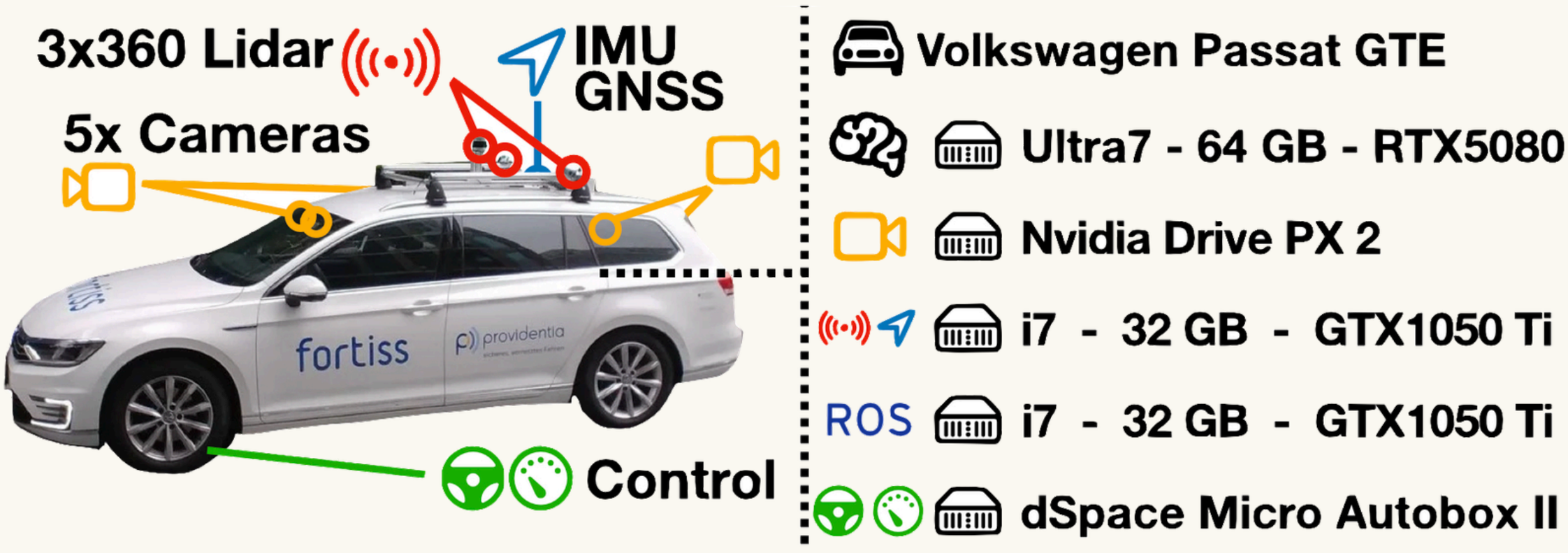
Sensors, compute, and control.

01 Cameras for visual perception and end-to-end driving

02 Three LiDARs for 360° geometry, obstacles, and modular stacks

03 RTK localization and onboard ROS compute

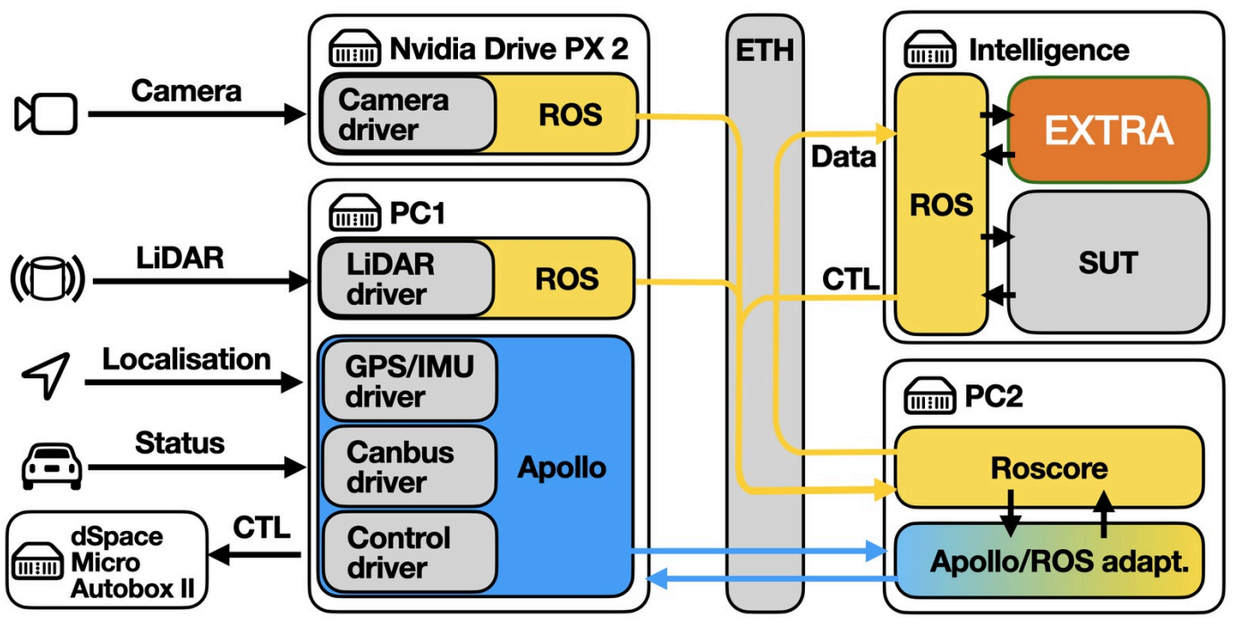
04 Real-time drive-by-wire actuation and control



HARDWARE AND LOGIC

Hardware and distributed computing pipeline.

The original platform ran on Apollo alone. I expanded it into a distributed system: camera and LiDAR drivers publish over ROS from an Nvidia Drive PX2 and a dedicated PC, Apollo still owns GPS/IMU, the CAN bus, and the real-time control interface that talks to the dSpace MicroAutoBox II, and everything meets over Ethernet. A bridge machine runs the ROS master alongside an Apollo/ROS adapter, so a separate "intelligence" node (the system under test, plus whatever extra tooling a given experiment needs) can subscribe to sensor data and publish control commands without touching Apollo directly.

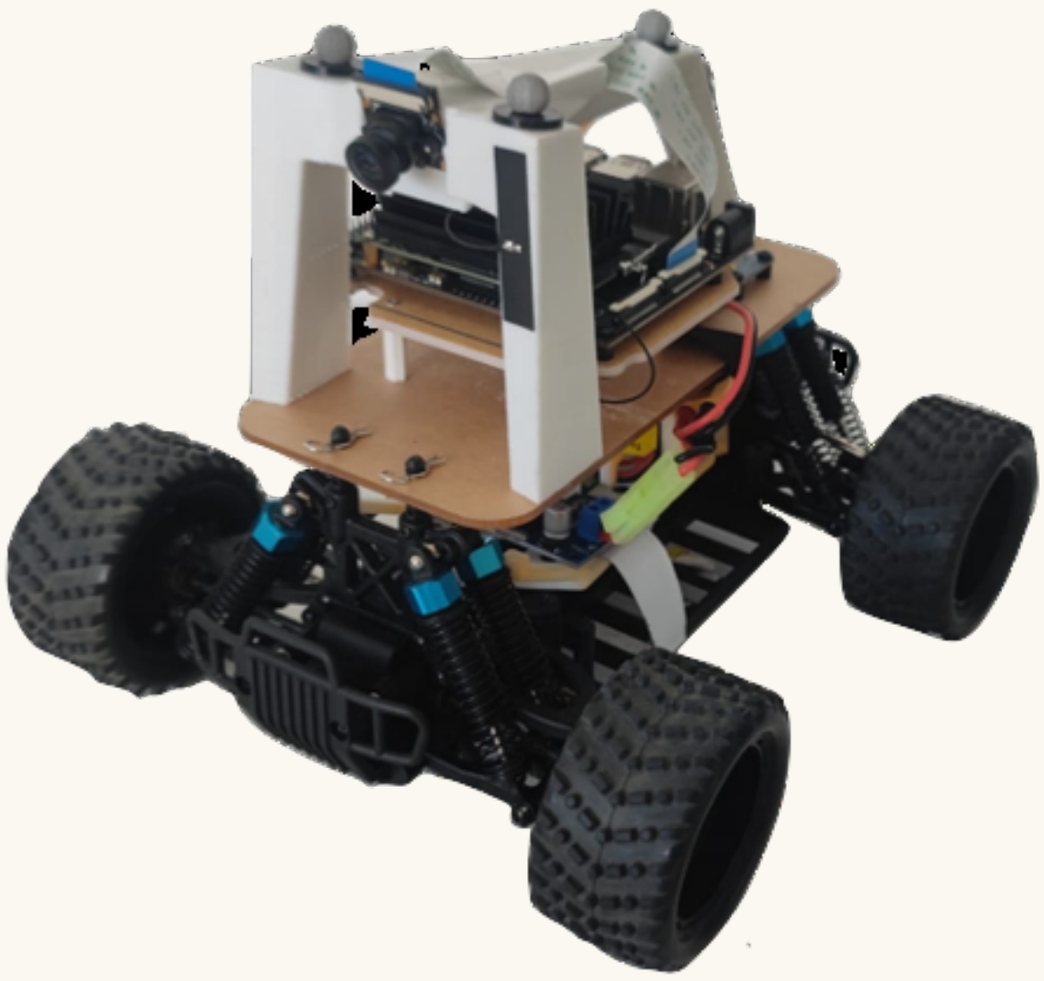


THE SMALL-SCALE PLATFORM

Donkey Car, Fortuna's *little brother.*

Fortuna has a smaller sibling: a 1/16-scale Donkey Car with its own small-scale lab and a mixed-reality framework built around a digital twin of that lab. For our ASE 2025 study, we used it to compare simulation, vehicle-in-the-loop, and mixed-reality testing against real driving—vehicle-in-the-loop reduced the actuation gap, and mixed-reality testing reduced the perception gap, compared to simulation alone.

READ THE ASE 2025 PAPER ↗





OPEN THE 3D MODEL ↗

JUST FOR FUN

I also scanned Fortuna into a 3D model.

You can take it for a lap around the track, no research question required.

OPEN IT UP ↗

GET IN TOUCH

SEND ME AN EMAIL ↗

GitHub ↗ Scholar ↗ DBLP ↗ LinkedIn ↗

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